

Uddhar

Open Data Sources for Earthquake Damage Triage in Nepal
Team YUKTI | Yantra Business Cup – SOFTBOTS AI Hackathon 2026

This reference lists the open datasets relevant to building and validating Uddhar. Note that most of the Uddhar pipeline is deterministic classical computer vision and requires no training data at all — co-registration, illumination normalisation, change detection, connected-component extraction, road-corridor analysis and zone scoring are all algorithmic. Only the severity classifier requires labelled examples. Everything else below serves as validation ground truth, raw imagery, or geometric input.

Priority markers: **[1]** obtain first — blocks core development. **[2]** needed for Nepal-specific validation. **[3]** useful, not blocking.

1. Labelled Training Data (severity classifier)

xBD / xView2 Building Damage Assessment Dataset [1]

The only large public dataset pairing pre- and post-disaster satellite imagery with building-level damage labels. Contains 22,068 images at 1024×1024 resolution across 19–22 disaster events in 15 countries, with roughly 850,000 human-annotated building polygons labelled on a four-point scale: no damage, minor, major, destroyed. Imagery is ~0.5 m resolution from the Maxar/DigitalGlobe open programme. This is the primary training source for the Uddhar severity classifier. For earthquake-specific calibration, prioritise the mexico-earthquake event; note that the dataset also contains a nepal-flooding event in the tier3 subset, useful for Nepali terrain familiarity though not for seismic damage signatures.

Challenge & download (registration required): <https://xview2.org/>

Dataset description: <https://xview2.org/dataset>

Paper (Gupta et al., 2019): <https://arxiv.org/abs/1911.09296>

Official baseline code: https://github.com/DIUX-xView/xView2_baseline

Hugging Face mirror (parquet): <https://huggingface.co/datasets/hannan022/xview2-xbd>

Size: ~30 GB | Format: PNG + GeoJSON labels | Licence: CC BY-NC-SA 4.0 (non-commercial)

KATE-CD — Kahramanmaraş Türkiye Earthquake Change Detection Dataset [3]

A focused earthquake change-detection dataset structured as pre-image / post-image / label triples at 512×512. Smaller than xBD but entirely earthquake-specific and drawn from Maxar Open Data and Airbus Pléiades imagery. Useful as a secondary calibration set precisely because the 2023 Türkiye earthquake damaged masonry building stock that resembles Nepali construction more closely than the timber-frame housing dominant in xBD.

Hugging Face: <https://huggingface.co/datasets/cscrs/kate-cd>

Format: paired images with masks | Also mirrored on GitHub and CodeOcean

2. Nepal Earthquake Ground Truth

2015 Nepal Earthquake: Open Data Portal (NPC / CBS / Kathmandu Living Labs) [1]

The most important Nepali dataset for this project. Following the 2015 Gorkha earthquake, the Government of Nepal conducted a census-style door-to-door survey across the 11 worst-affected districts, deploying 3,000 engineers with tablets who collected 10 TB of data and 10 million photographs in just over 120 days. The final dataset covers 762,106 buildings and 3.6 million individuals, with each building assigned an ordinal damage grade — from hairline cracks and non-structural damage at Grade 1 through to complete collapse — alongside construction type, foundation, superstructure material, storey count and socio-economic attributes.

Portal: <https://eq2015.npc.gov.np/>

Download: <https://eq2015.npc.gov.np/download>

API documentation: <https://eq2015.npc.gov.np/docs>

Kathmandu Living Labs project page: <http://kathmandulivinglabs.org/our-work/earthquake-data-portal>

Format: CSV / API | Licence: Open Database License (ODbL) | Owner: National Planning Commission

Important: this is tabular survey data, not labelled imagery, and records are anonymised to administrative units rather than carrying per-building coordinates. It therefore cannot train a pixel-level classifier. It is, however, close to ideal ground truth for the metric that actually matters to Uddhar — zone-level ranking. Aggregating damage grades by ward gives a true severity ordering that predicted zone rankings can be scored against, producing a top-k recall figure

validated on real Government of Nepal data.

NDRRMA BIPAD Portal — Disaster Information Management System [2]

Nepal's national disaster information platform, owned by NDRRMA under the Ministry of Home Affairs, with separate national, provincial and municipal instances. Holds incident records reported by Nepal Police dating from 2011 — including deaths, injuries, missing persons, infrastructure destroyed and economic loss — plus modules for risk information, damage and loss data, and impact-based forecasting built on OpenStreetMap building and road layers. Serves two roles for Uddhar: a validation source for recent events, and the export schema target for GeoJSON output.

BIPAD Portal: <https://bipadportal.gov.np>

BIPAD information site: <https://www.bipad.gov.np/>

Nepal DRR Portal: <http://drportal.gov.np/>

Format: web portal with map/table views | Owner: Government of Nepal, NDRRMA

3. Satellite and Aerial Imagery (pre/post pairs)

OpenAerialMap (HOT / Development Seed) [1]

An open catalogue of satellite and UAV imagery, all openly licensed under CC BY 4.0. Uniquely relevant here because it was founded partly on the Nepal earthquake response and holds imagery from that operation. It is also the single best source of drone imagery generally, which matters because Uddhar is designed drone-first: UAVs fly beneath cloud, need no satellite tasking request and deliver resolution far exceeding commercial satellite. Pan to Gorkha, Sindhupalchok, Nuwakot or Dhading and filter by date to locate 2015 coverage.

Imagery browser: <https://map.openaerialmap.org/>

AWS Open Data registry: <https://registry.opendata.aws/openaerialmap/>

Programme information: <https://www.hotosm.org/en/tools-resources/tech-product-suite/open-aerial-map/>

Format: Cloud-Optimized GeoTIFF | Licence: CC BY 4.0 | API available

Maxar (Vantor) Open Data Program [1]

Free pre- and post-event high-resolution satellite imagery released in response to disaster activations. Approximately 150,000 images totalling around 14 TB at 30–50 cm resolution, organised by event and supplied as analysis-ready Cloud-Optimized GeoTIFFs. This is the source from which xBD imagery was drawn. Check the event list for Nepal 2015 coverage; the QGIS plugin is by far the easiest entry point, allowing you to browse events, preview footprints, swipe-compare pre and post imagery and download without an AWS account or any code.

Programme and event list: <https://www.maxar.com/open-data>

AWS Open Data registry: <https://registry.opendata.aws/maxar-open-data/>

STAC catalog (JSON): <https://maxar-opendata.s3.amazonaws.com/events/catalog.json>

Python-friendly catalog (GeoJSON / CSV): <https://github.com/opengeos/maxar-open-data>

QGIS plugin — recommended starting point: https://plugins.qgis.org/plugins/maxar_open_data/

Source Cooperative mirror: <https://source.coop/maxar/maxar-opendata>

Format: COG | Resolution: 30–50 cm | Free, no account needed via QGIS plugin

Copernicus Sentinel-2 (ESA) [3]

Free multispectral optical imagery with a five-day revisit cycle at 10 m resolution. Too coarse to resolve individual Nepali houses, so unsuitable as a primary input for building-level damage assessment. Still useful for regional context, landslide-scar mapping, wide-area change screening and for establishing where higher-resolution tasking should be directed.

Copernicus Data Space: <https://dataspace.copernicus.eu/>

USGS EarthExplorer: <https://earthexplorer.usgs.gov/>

Resolution: 10 m | Revisit: ~5 days | Licence: free and open

4. Terrain Data (required for slope-aspect normalisation)

A digital elevation model is a hard dependency for Uddhar's core technical contribution. Slope and aspect cannot be computed without one, and without slope-aspect banding the illumination normaliser will misread entire hillside faces as structural change. Obtain a DEM before implementing that stage.

Copernicus DEM GLO-30 [1]

A 30 m global digital elevation model, generally regarded as the highest-quality free DEM currently available and noticeably better than SRTM in steep terrain. Recommended default for computing slope and aspect across Nepali hill districts.

AWS Open Data registry: <https://registry.opendata.aws/copernicus-dem/>

OpenTopography portal: <https://portal.opentopography.org/>

Resolution: 30 m | Format: GeoTIFF | Free

NASA SRTM / ALOS World 3D [3]

Established 30 m global elevation models. Useful as cross-checks against Copernicus DEM, or as fallbacks where Copernicus coverage has gaps. Both are well documented and widely used in Himalayan terrain analysis.

USGS EarthExplorer: <https://earthexplorer.usgs.gov/>

OpenTopography portal: <https://portal.opentopography.org/>

Resolution: 30 m | Format: GeoTIFF | Free

5. Road and Building Geometry

OpenStreetMap Nepal — Geofabrik extract [1]

Daily-updated extract of all OpenStreetMap data for Nepal, including road centrelines, building footprints and administrative boundary relations. Road geometry is the direct input to Uddhar's blockage-inference stage: corridors are buffered from these centrelines, change statistics are computed along them, and the resulting traversability states rebuild the routing graph. Nepal's OSM coverage is unusually good because more than 7,000 volunteers mapped roads, buildings, IDP camps and helicopter landing sites during the 2015 earthquake response.

Nepal extract: <https://download.geofabrik.de/asia/nepal.html>

HOT Export Tool (custom area / tags): <https://export.hotosm.org/>

2015 Nepal earthquake HOT activation record: https://wiki.openstreetmap.org/wiki/2015_Nepal_earthquake

Format: PBF / Shapefile | Licence: ODbL (share-alike applies to derived data)

Google Open Buildings [2]

Machine-derived building footprints for the Global South, covering areas where OpenStreetMap building coverage is sparse — which describes much of rural Karnali and the far-western hills. Provides pre-event structure priors so that damage can be expressed as a proportion of buildings affected rather than as a raw count. This is one of the two models underpinning the UNOSAT–Google DISHA damage assessment workflow.

Dataset and download: <https://sites.research.google/open-buildings/>

Format: CSV / GeoJSON polygons | Licence: CC BY 4.0 / ODbL

Nepal Administrative Boundaries (palika and ward) [2]

Local government and ward boundary geometry, used to snap Uddhar's analysis grid to real jurisdictions. This matters operationally: a zone that maps to an actual ward maps to an accountable authority who can be contacted and tasked, whereas an arbitrary grid cell does not. Boundary relations are available within the Geofabrik OSM extract, and also through HDX.

Humanitarian Data Exchange — Nepal: <https://data.humdata.org/group/npl>

Geofabrik Nepal extract (admin relations): <https://download.geofabrik.de/asia/nepal.html>

Format: Shapefile / GeoJSON | Licence: varies by source, generally open

6. Reference Models and Prior Art

SKAI — WFP Innovation Accelerator and Google Research [3]

An open-sourced tool using AI and satellite imagery to assess building damage automatically at scale, co-developed by the World Food Programme's Innovation Accelerator and Google Research. Worth studying as the closest open equivalent to the deep-learning approach Uddhar deliberately avoids, and useful for framing the comparison in Section 5 of the proposal.

Project page: <https://innovation.wfp.org/project/SKAI>

Open source | Requires GPU infrastructure

xView2 baseline and published solutions [3]

Reference implementations and reported scores against which Uddhar's classical pipeline can be positioned. The official challenge metric is a weighted mean of 30 per cent localisation F1 and 70 per cent damage F1, which gives a directly comparable benchmark figure for the proposal.

Official baseline: https://github.com/DIUX-xView/xView2_baseline

Strong simple baseline (paper): <https://arxiv.org/abs/2401.17271>

Use for benchmarking context, not as a component

7. Recommended Acquisition Order

Step	Dataset	Unblocks
1	xBD / xView2	Severity classifier training and benchmark accuracy figures
2	Copernicus DEM GLO-30	Slope-aspect illumination normalisation — the core innovation
3	OSM Nepal (Geofabrik)	Road-corridor blockage inference and access-time routing
4	OpenAerialMap	Real Nepali pre/post imagery for domain validation
5	eq2015 NPC portal	Ward-level ground truth for zone-ranking accuracy
6	Maxar Open Data	Additional pre/post pairs; higher-resolution alternatives
7	BIPAD portal	Export schema definition and recent-event validation

8. Notes on Gaps and Licensing

No open imagery is expected for the November 2023 Jajarkot earthquake. Events at that scale generally fall below the threshold that triggers large open-imagery releases. If a search of the Maxar event list and OpenAerialMap returns nothing, record that result rather than treating it as a setback: an absence of open imagery for Nepal's most recent damaging earthquake is direct evidence for the central argument of the proposal — that domestically operable, drone-capable tooling is necessary because international imagery activation does not reach events of this size.

Fallback validation if no Nepali pre/post pair can be obtained. The slope-shadow problem can still be demonstrated using two clear-sky acquisitions of an undamaged Nepali hillside taken at different sun angles. If the normaliser reports near-zero change across that pair while a global histogram-equalisation baseline reports substantial false change, the core technical contribution is proven with evidence rather than asserted.

Licence obligations to observe. xBD is CC BY-NC-SA 4.0 and therefore non-commercial, which is acceptable for the hackathon and for research but relevant to the commercial tiers described in the business model. OpenStreetMap is ODbL, whose share-alike provision extends to derived data — the GeoJSON export will inherit it. OpenAerialMap is CC BY 4.0 and requires attribution to contributors of the Open Imagery Network. The eq2015 portal is ODbL. Attribution statements for each should appear in the application interface, not only in documentation.

Uddhar — Access-Aware Earthquake Damage Triage for Nepal. All URLs verified at time of compilation; open datasets change, so confirm availability before relying on any single source.